

MAKSIM ZHDANOV

email · website · github · google scholar · twitter · linkedin

research: sparse attention; long-context modeling; hardware-aligned architectures;
large-scale weather forecasting

EDUCATION

PhD in Machine Learning 2023 – 2027

University of Amsterdam, AMLab

· Advisors: Max Welling & Jan-Willem van de Meent

MSc in Computer Science 2019 – 2022

TU Dresden (GPA 1.4, excellent)

BSc in Physics 2015 – 2019

Saint Petersburg State University (GPA 4.8/5.0, with honours)

EXPERIENCE

Research Intern Jun 2026 – present

Google DeepMind, London

· Weather team, hosted by Ferran Alet & Ilan Price

· Hardware-aligned architectures for large-scale weather forecasting

Research Intern Feb 2026 – May 2026

CuspAI, Amsterdam

· Hosted by Govert Verkes

· Agentic search for materials discovery

Research Assistant Sep 2020 – Apr 2023

Helmholtz AI, Dresden

· ML for neuroscience and materials science

PUBLICATIONS

Native Multi-Dimensional Subquadratic Operators via Input Dependent Long Convolutions

David R. Wessels, Farhad Ramezanghorbani, Alireza Moradzadeh, David W. Romero, Olivia Viessmann, [Maksim Zhdanov](#), John St. John, Ken Janik, David M. Knigge, Yucheng Tang, Erik J. Bekkers, Saeed Gopal Paliwal

NeurIPS 2026

[arxiv](#)

Follow the Mean: Reference-Guided Flow Matching

Pedro M. P. Curvo, [Maksim Zhdanov](#), Floor Eijkelboom, Jan-Willem van de Meent

NeurIPS 2026

[arxiv](#)

(Sparse) Attention to the Details: Preserving Spectral Fidelity in ML-based Weather Forecasting Models

[Maksim Zhdanov](#), Ana Lucic, Max Welling, Jan-Willem van de Meent

ICML 2026; Best Paper Runner-up, FM4S @ ICLR 2026

[arxiv](#)

[weights](#)

[demo](#)

Conditional Clifford-Steerable CNNs with Complete Kernel Basis for PDE Modeling

Bálint László Szarvas*, [Maksim Zhdanov](#)*

ICML 2026

[arxiv](#)

[code](#)

MSPT: Efficient Large-Scale Physical Modeling via Parallelized Multi-Scale Attention

Pedro Curvo, Jan-Willem van de Meent, [Maksim Zhdanov](#)

CVPR 2026 (Highlight)

[arxiv](#)

[code](#)

AdS-GNN: a Conformally Equivariant Graph Neural Network

Maksim Zhdanov, Nabil Iqbal, Erik Bekkers, Patrick Forré

ICLR 2026

[arxiv](#) [code](#)

Adaptive Mesh Quantization for Neural PDE Solvers

Winfried van den Dool*, Maksim Zhdanov*, Yuki Asano, Max Welling

TMLR 2025

[arxiv](#) [code](#)

Erwin: A Tree-based Hierarchical Transformer for Large-scale Physical Systems

Maksim Zhdanov, Max Welling, Jan-Willem van de Meent

ICML 2025

[arxiv](#) [code](#) [blog](#)

Clifford-Steerable Convolutional Neural Networks

Maksim Zhdanov, David Ruhe, Maurice Weiler, Ana Lucic, Johannes Brandstetter, Patrick Forré

ICML 2024

[arxiv](#) [code](#) [blog](#)

Implicit Convolutional Kernels for Steerable CNNs

Maksim Zhdanov, Nico Hoffmann, Gabriele Cesa

NeurIPS 2023

[arxiv](#) [code](#) [blog](#)

Investigating Brain Connectivity with Graph Neural Networks and GNNExplainer

Maksim Zhdanov, Saskia Steinmann, Nico Hoffmann

ICPR 2022 (Oral)

[arxiv](#) [code](#)

* equal contribution

WORKSHOP PAPERS

BSA: Ball Sparse Attention for Large-scale Geometries

Catalin E. Brita, Hieu Nguyen, Lohithsai Yadala Chanchu, Domonkos Nagy, Maksim Zhdanov

LCFM @ ICML 2025

[arxiv](#) [code](#)

Amortized Bayesian Inference of GISAXS Data with Normalizing Flows

Maksim Zhdanov, Lisa Randolph, Thomas Kluge, Motoaki Nakatsutsumi, Christian Gutt, Marina Ganeva, Nico Hoffmann

ML4PS @ NeurIPS 2022

[arxiv](#)

Learning Generative Factors of EEG Data with Variational Auto-Encoders

Maksim Zhdanov, Saskia Steinmann, Nico Hoffmann

DGM4MICCAI 2022

[arxiv](#)

INVITED TALKS

Triton & Flash Attention Workshop, ELLIS Deep Thinking Hour

Dec 2025

AI4Science Reading Group, Mila, Montréal

Aug 2025

TEACHING

Teaching Assistant, University of Amsterdam

2023 – 2025

· Machine Learning I (2023); Deep Learning II (2024, 2025)

· Guest lecture: Introduction to Triton, kernel fusion and Flash Attention (2026)

TECHNICAL SKILLS

Code: Python, C++, Triton

ML: JAX, PyTorch; large-scale TPU/GPU training